Filtering Through the Storm

Discussion of

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CEMLA Conference, Mexico City

November 5, 2026

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Main Idea

- How should a DSGE model handle episodes of extreme volatility?
- Authors conduct a horse race of four treatments, in Colombian model (CUMBIA)
  1. Constant variance shocks.
  2. Treat real activity variables as missing during volatile episodes
  3. Add episode-specific AR(1) processes
  4. Add episode-specific white noise
- Conclude T3 forecasts best post-COVID; also favor T4 for implementation/explanatory ease

Difference between Treatments 3 and 4 (paper's equations)

Let $x_t \equiv \zeta_t - \bar{\zeta}$:

Treatment 3, Eq. (4.1): $x_t = \rho x_{t-1} + \varepsilon_t + \varepsilon_t^H$

Treatment 4, Eq. (4.3): $x_t = \rho x_{t-1} + \varepsilon_t + \varepsilon_t^H - \rho \varepsilon_{t-1}^H$

Difference between Treatments 3 and 4 (lag operator notation)

Lag operator $Lx_t = x_{t-1}$:

Treatment 3: $(1 - \rho L)x_t = \varepsilon_t + \varepsilon_t^H$

Treatment 4: $(1 - \rho L)x_t = \varepsilon_t + (1 - \rho L)\varepsilon_t^H$

Difference between Treatments 3 and 4 (invert)

For $|\rho| < 1$,

$$\text{Treatment 3:} \qquad x_t = \frac{1}{1-\rho L}\varepsilon_t + \frac{1}{1-\rho L}\varepsilon_t^H$$

$$\text{Treatment 4:} \qquad x_t = \frac{1}{1-\rho L}\varepsilon_t + \varepsilon_t^H$$

Additional term is AR(1) with same ρ under T3; white noise under T4.

Major Comment 1: Real-time Forecasting

The current exercise is "pseudo out-of-sample" in that:

- Each date's forecast Kalman filters using data up to that date
- ... but uses the parameter values from the entire 2000Q1–2025Q4 sample's MLE.
- Especially problematic for forecasting at the beginning of an episode

I recommend a more complete out-of-sample test, with the model re-estimated at each date, using only data available up to that point.

Related: forecast rankings are period-dependent

Average RMSE improvement relative to Treatment 1, across GDP growth, inflation, and the policy rate:

	Pre-COVID	During COVID	Post-COVID
Treatment 2: omit real data	−6.6%	−15.2%	−8.6%
Treatment 3: persistent	−2.3%	−9.1%	+12.8%
Treatment 4: transitory	−5.1%	−11.2%	+6.9%

T1 is best before/during COVID, T3 is best afterwards. Authors should make clear why they care more about the latter. Say both:

- A protocol for "practitioners facing the next such episode"
- but evaluate "which one forecasts better once normal times resume"

Major comment 2: What exactly is T4's advantage?

Treatment 3 forecasts better than Treatment 4 in every row of Table 6.1. So why argue for using both? Need to clarify:

- T4 is easier to explain to policy makers?
- T4 is easier to implement?
  - Authors mention some concerns about collinearity in T3
  - So, is it identified or isn't it?
- Explain the choice of $s_\tau = (2, 2, 3)$ and show forecast sensitivity to it. What operational benefit makes Treatment 4 worth its lower forecast accuracy?

Major comment 3: Why Treatment 2?

- Plenty of citations for the other methods; do people actually use T2?
- Clarify: Why is it a reasonable idea to discard several time series?
- .... and why these seven: GDP, consumption, investment, exports, imports, foreign GDP, and trend growth

Ideas for increasing potential impact

Practioners will wonder: do the CUMBRIA conclusions generalize?

- Repeat the horse race in other DSGE models (many official central bank models are available)
- Can we expand the horse race? i.e. other approaches for non-Gaussianity:
  - peristent stochastic volatility
  - fat-tailed innovations
- Extended sample? 100 quarters + 27 observables for 50+ paramters would beenfit from more data
- Additional settings? CUMBRIA appears general enough to apply to other small open economies.

Extra thoughts

- Non-Gaussianity vs non-linearity
- AI use: writing (clarity vs strategy), ambition